# ELI — Diagrams

ELI v2.5.11

October 2026

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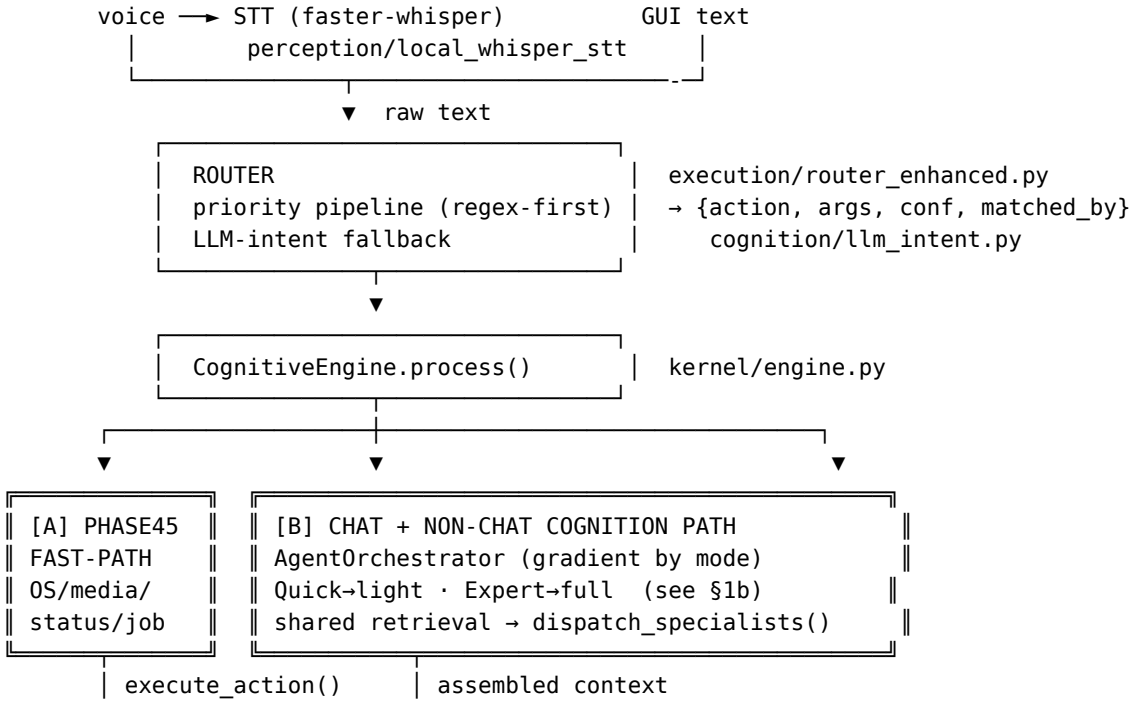
## Blueprint — ELI ASCII Diagrams

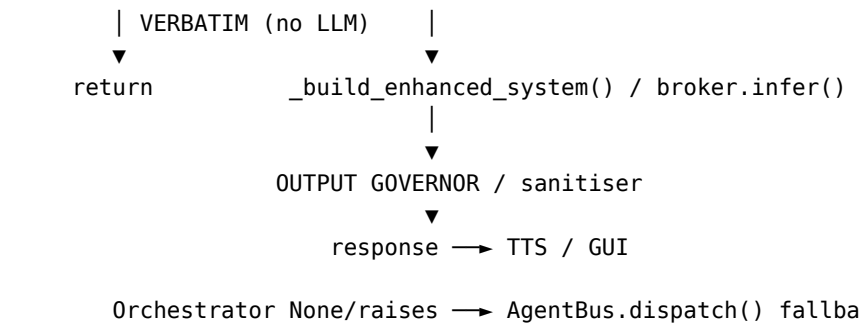
> **Updated for v2.4.84.** CHAT uses a gradient orchestrator for all modes; retrieval is unified in `eli/memory/retrieval.py`; Stage 12 = learning/state commit.

Visual companion to `architecture.md`. Three views: the full request pipeline, the memory subsystem, and the gating stack. All grounded in the real modules (file paths are clickable).

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### 1. Full pipeline (request lifecycle)



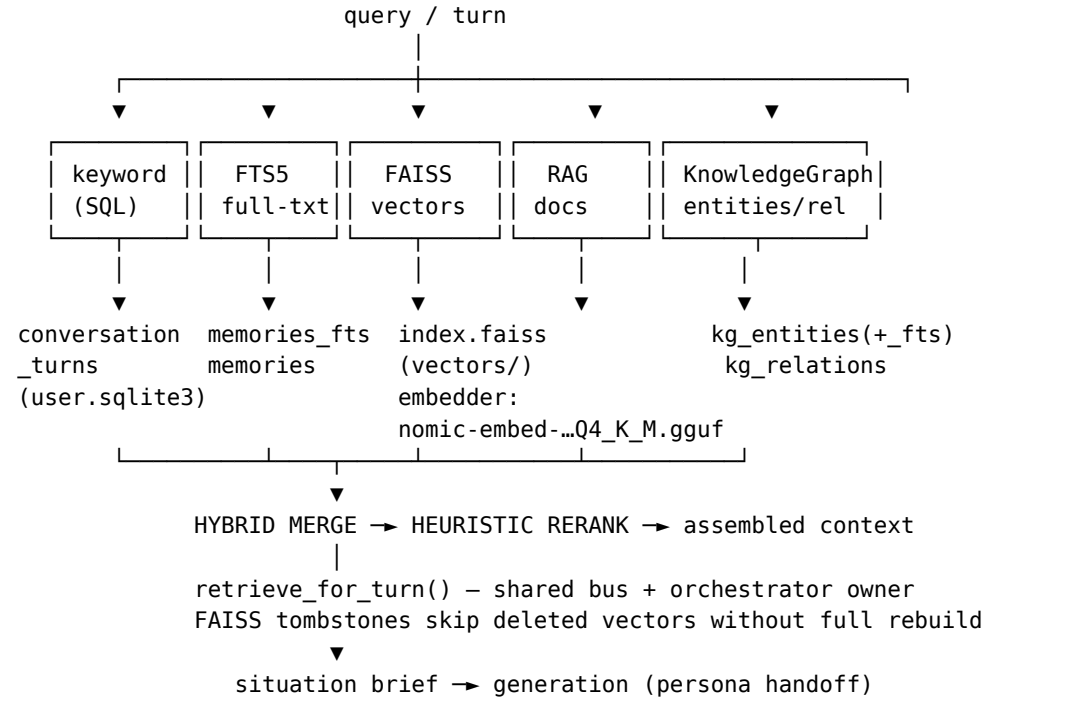



### 1b. Orchestrator — canonical S01–S12 (`pipeline_trace.py`)

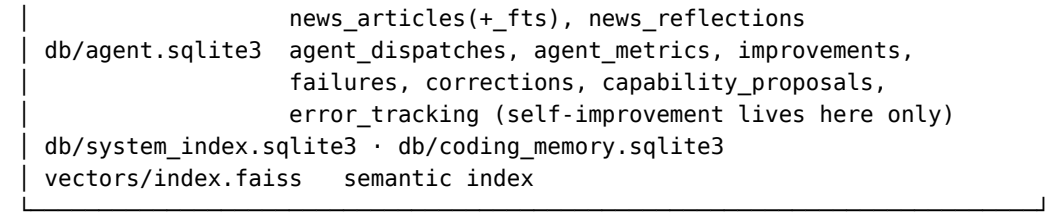
```
S01 PERCEIVE → S02 INPUT_GUARDS → S03 ROUTER → S04 GROUNDING_GATE
       → S05 PLANNER (mode-aware budgets)
       → S06 AGENT_BUS: retrieve_for_turn() + dispatch_specialists()
       → S07 CONTEXT_ASSEMBLY (merge + heuristic rerank)
       → S08 INFERENCE_BROKER → S09 REASONING_SYNTHESIS (mode algo)
       → S10 OUTPUT_GOVERNOR → S11 RESPONSE_DELIVERY
       → S12 LEARNING_STATE_UPDATE (learning_coordinator.finalize_turn)
```

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## 2. Memory subsystem `eli/memory` + `cognition/working_memory.py`




```
┌──────────────────────── STORES (artifacts/) ─────────────────────────┐
│ db/user.sqlite3   memories, memories_archive, memory_meta, memory_lineage,│
│                   semantic,                                          │
│                   conversation_turns, conversations, session_summaries,   │
│                   kg_entities(+_fts), kg_relations, recall_log,           │
│                   runtime_events, learning_replay, observations,          │
│                   habits/habit_events/habit_rules, user_patterns,         │
│                   user_model, eli_stances, belief_revisions,              │
```

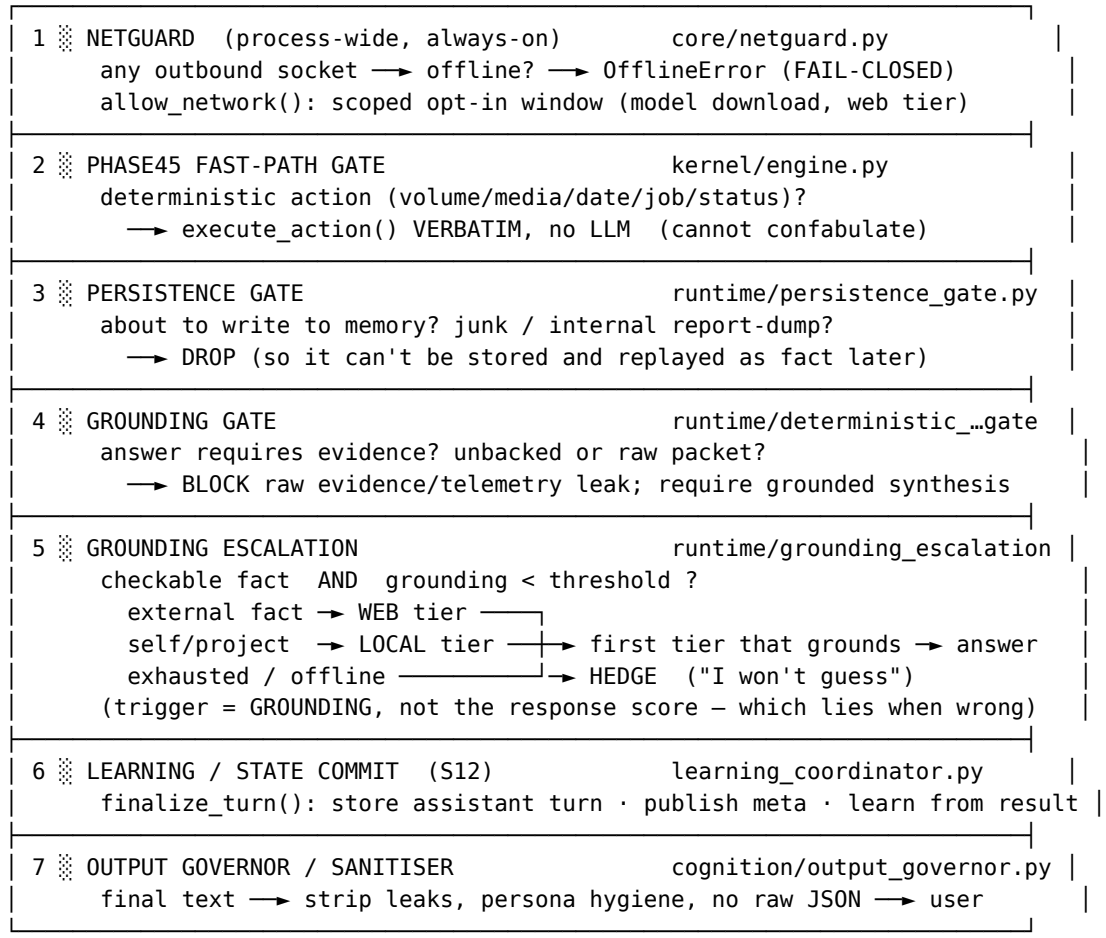



A question that names a period ("last week") is filtered by date FIRST (Memory.memories_between + dated conversation turns), then ranked by topic.

WRITE PATH: turn → PERSISTENCE GATE (drop junk/report-dumps) → STORAGE POLICY (origin · dedupe · importance) → store; daily upkeep decays and archives

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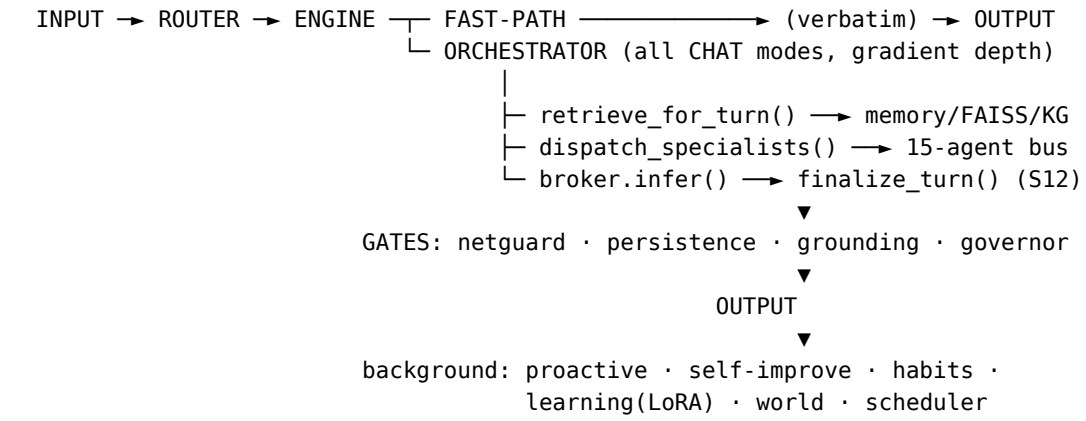
## 3. Gating stack (in path order — every guard the request passes)




Legend: FAIL-CLOSED = denies by default; VERBATIM = returned without an LLM pass; HEDGE = honest "can't verify" instead of a guess.

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## 4. One-screen overview (how it all connects)




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